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Geological Society of America Bulletin

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Geological Society of America Bulletin 1992;104;1301-1314
doi: 10.1130/0016-7606(1992)104<1301:OOTCAO>2.3.CO;2

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Origin of the Central American ophiolites: Evidence from paleomagnetic results

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ABSTRACT

The ophiolite complexes exposed along the Pacific coast of Costa Rica and western Panama evolved from Jurassic ocean floor to the island arc of the Tertiary to the present. We attribute Late Cretaceous volcanism to the Caribbean sill event that thickened the oceanic crust to form an oceanic plateau. Tectonostratigraphic units formed by Late Cretaceous nappe thrusting and block rotation. Rotated blocks, as revealed by paleomagnetic data, are covered by neoautochthonous sedimentary and volcanic rocks. Collision with the Cocos Ridge in the late Tertiary fragmented and rotated the ophiolite complexes south of the Costa Rica Fracture Zone.

The oceanic basement formed in the Jurassic Period in an equatorial position. Its latitudinal drift path as deduced from paleomagnetic inclination data matches that of the South American continent and differs from that of the Farallon/Phoenix plates. We favor formation of the ophiolite complexes in a Caribbean (inter-American) position as opposed to a distant origin in the Pacific region.

INTRODUCTION

Ophiolite complexes line up along the Pacific coast of Costa Rica, Panama, and Colombia (Fig. 1). They have a complex history from Jurassic through Paleocene time and probably were derived from an originally coherent body (Pichler and others, 1974; Goosens and others, 1977; Schmidt-Effing, 1979). Their origin is a matter of debate. Most authors assume that they represent ocean floor that formed in the Pacific region in the Jurassic and drifted from there into its present position (for example, Wilson, 1965; Burke and others, 1978; Duncan and Hargraves,1984; Pindell and others, 1988; Ross and Scotese, 1988). Wildberg and others (1981), however, considered the Costa Rican ophiolites as an *in situ*–formed part of the later Caribbean plate. Seely and others (1974) and Seely (1979) interpreted the Nicoya Complex in northwestern Costa Rica as an accretionary prism characterized by thrust faults along a convergent and shortened plate margin.

Only scarce paleomagnetic data from Costa Rica and Panama have been published to date. De Boer (1979) presented data for 87 samples from basaltic rocks of the Nicoya peninsula without information on the number of sample sites or locations. He inferred gradual clockwise rotation from an east-west–trending aeromagnetic pattern in the northern Nicoya peninsula, although the paleomagnetic data do not support such an interpretation. Gose (1983) carried out a paleomagnetic study on post-Campanian sedimentary rocks. His results, obtained from 324 samples of 14 sites, reveal formation of these rocks in an equatorial position and post-Eocene northward translation with no rotation.

Case and others (1984) and Howell and others (1985) defined the Nicoya Complex as a suspect tectonostratigraphic terrane. The place of origin and mode of emplacement remain obscure. It is the goal of this paper to examine whether the concept of suspect allochthonous terranes, such as exposed along the western margin of the North American plate (Ben-Avraham and others, 1981), is applicable to the ophiolite complexes of the Central American landbridge and whether the place of origin is the Pacific or rather the Caribbean region in geographic terms. Our geochemical and paleomagnetic studies show that the ophiolite bodies along the Pacific coast of Costa Rica and Panama contain basalts with different chemical characteristics and that they underwent complex block rotation. On the basis of these data, we present a plate-tectonic model for the evolution of the Central American landbridge.

THE OPHIOLITE COMPLEXES NORTH OF THE COSTA RICA FRACTURE ZONE

One of the largest and best studied ophiolite exposures is the Nicoya Complex on the Nicoya peninsula in northwestern Costa Rica (Romanes, 1912; Dengo, 1962). Wildberg and others (1981) and Gursky and others (1984) subdivided the Nicoya Complex into two lithostratigraphic subcomplexes, the Lower and the Upper Nicoya Complex (Fig. 2). The Lower Nicoya Complex contains the Santa Elena ultramafic body, mid-ocean ridge basalt (Jurassic), and deep-sea sedimentary rocks (mainly radiolarian chert; Jurassic-Cretaceous). The Upper Nicoya Complex consists mainly of a late Cretaceous basaltic suite.

Based on observations of Kuijpers (1980) and Tournon and Azéma (1980), Bourgeois and others (1984) presented a tectonic model with three nappes in the Nicoya peninsula and the neighboring Santa Elena peninsula (Fig. 3). Both the lower Esperanza and the higher Matapalo Nappe contain basaltic rocks and radiolarian chert ("volcanic-sedimentary unit" of Tournon, 1984). The Santa Elena Nappe, the tectonically highest unit, consists mainly of ultramafic rocks. The Santa Elena and Matapalo nappes are essentially made up of rocks from the Lower Nicoya Complex. The Esperanza Nappe contains the sequence of the Upper Nicoya Complex.

The Rock Sequence

The *Santa Elena Nappe* consists of harzburgite and dunite partly serpentinized, gabbro, amphibolite, and dolerite dikes (Tournon and Azéma, 1980). It represents the deepest exposed level of the ophiolite sequences, namely uppermost mantle and lower crust. The harzburg-

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Geological Society of America Bulletin, v. 104, p. 1301–1314, 12 figs., 2 tables, October 1992.

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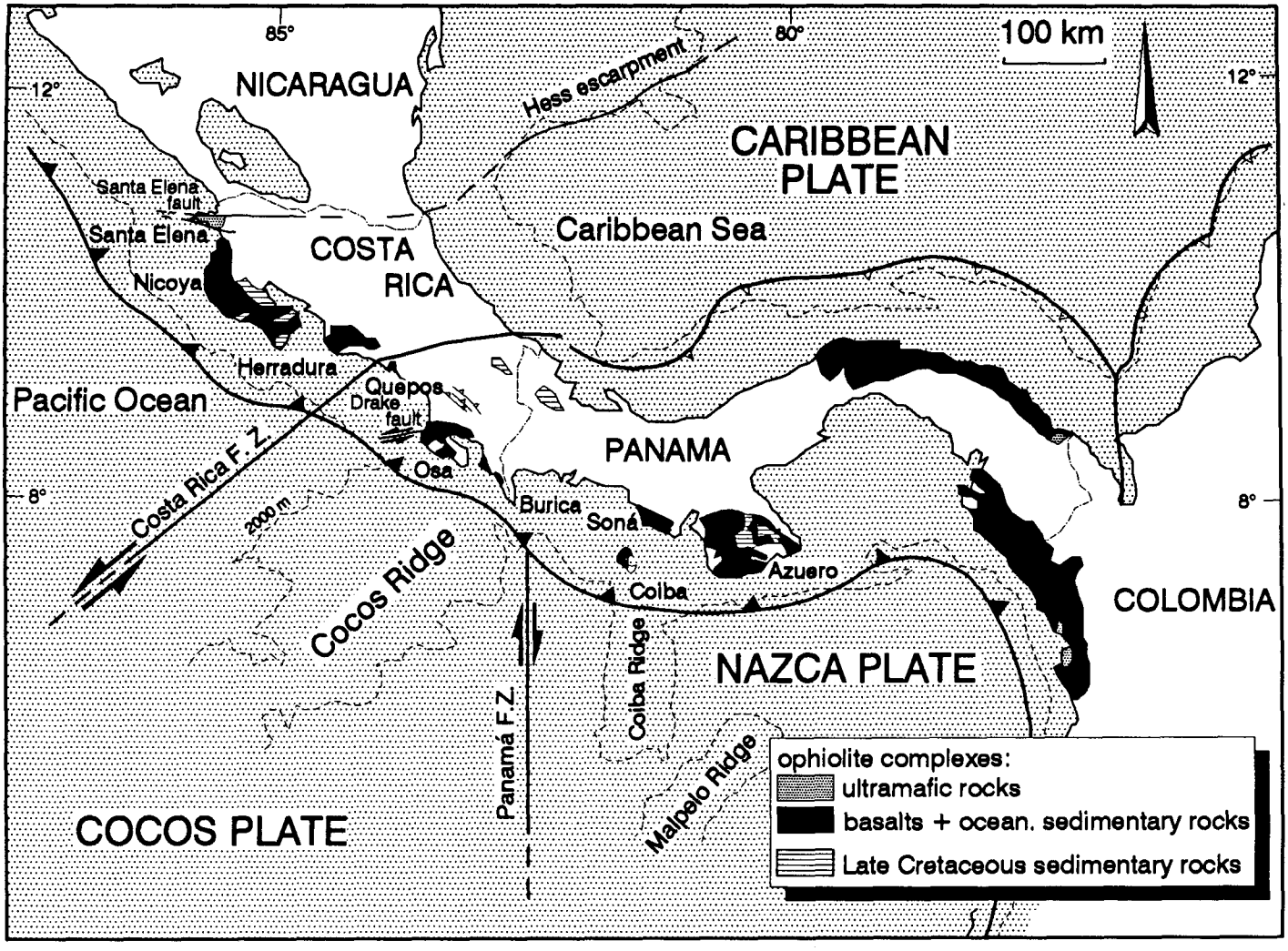


Figure 1. Ophiolite complexes and main tectonic elements in Central America (modified after Berrangé and Thorpe, 1988). F.Z., fracture zone.

ite contains dunite bodies of varying size with podiform and disseminated chromite mineralization (Kuijpers and Jäger, 1979). Layered peridotite and gabbro are interpreted as cumulates formed at the base of the oceanic crust (Desmet and Rocci, 1988). Other gabbro bodies are not layered.

Amphibolite, which forms a sliver near the base of the nappe, is ductilely deformed and has preferred mineral orientation, indicating that the base of the nappe was hot during emplacement. K/Ar dating on amphibole yielded a metamorphic cooling age of 89 Ma (Bellon and Tournon, 1978). The underlying basalt and sedimentary rocks of the Matapalo Nappe are not metamorphosed. Tectonic flow criteria (S-C fabrics; Berthé and others, 1979) from mylonitized serpentinite along the sole of the Santa Elena Nappe indicate a south- to west-southwest–directed emplacement (Fig. 4). Preferred orientation of the long axes of hornblende in the amphibolites and fold polarity in the sedimentary sequence of the overthrust unit (see also Tournon and Azéma, 1980) are consistent with these data.

The *Matapalo Nappe* consists mainly of pillow and flow basalts representing the oceanic basement. The basalts are associated with plagiogranite and diorite as well as pelagic sedimentary rocks. The Matapalo Nappe is exposed in the northern part of the Nicoya peninsula and the Santa Elena peninsula (Fig. 3, 4).

The sedimentary cover of the Matapalo Nappe is composed of radiolarian chert and intercalated breccia, and contains dolerite sills and dikes. Its contacts with the basement are all tectonic. Incompetent claystone layers caused offscraping and enabled disharmonic folding of the well-stratified sedimentary sequence. The sedimentary rocks range in age from Jurassic to Santonian (Schmidt-Effing, 1979; Tournon, 1984; DeWever and others, 1985; Baumgartner, 1987). The breccias, exposed in the Santa Elena peninsula, contain radiolarian chert as the main component and have graded bedding. The oldest fossils are Early Jurassic (DeWever and others, 1985). It is not clear, however, whether the matrix or the detrital components contain these fossils. The oldest dated radiolarian chert is Middle Jurassic (Schmidt-Effing, 1979; Baumgartner, 1987).

The *Esperanza Nappe* occupies most of the Nicoya peninsula and is also exposed in Herradura (see Fig. 3). It consists of pillow and flow basalts intruded by dolerite dikes and sills. A pillowed basalt from the Nicoya peninsula yielded a K/Ar age of 85.5 (Santonian) ± 4.7 Ma (Barr and Escalante, 1969; recalculated with $\lambda_K = 4.962 \times 10^{-10} a^{-1}$).

After nappe thrusting and folding in the Late

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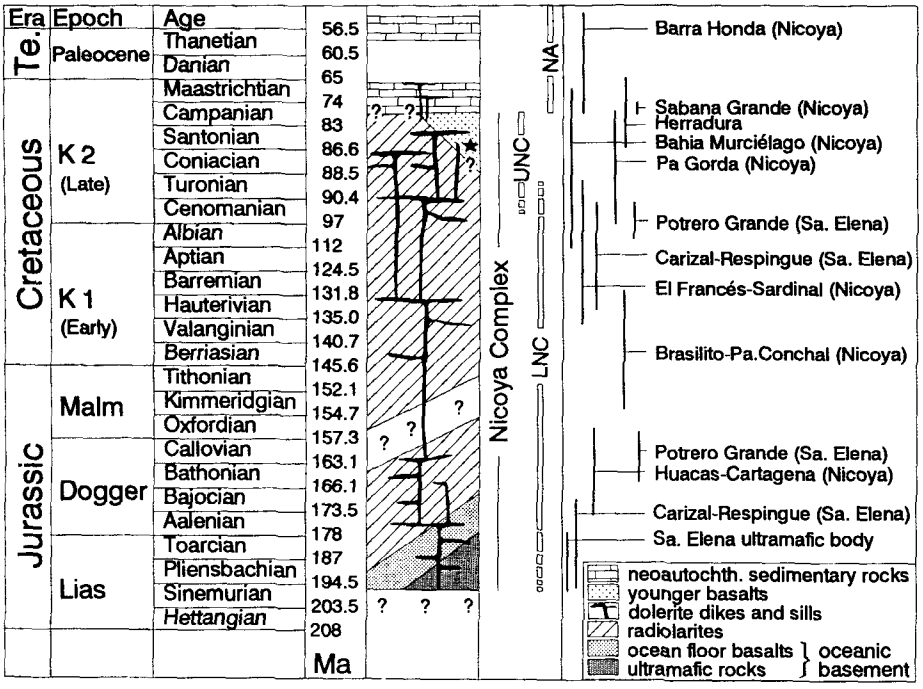


Figure 2. Composite stratigraphic column of the ophiolite complexes in northern Costa Rica, based on biostratigraphic dating after Baumgartner (1987), Schmidt-Effing (1979), and Tournon (1984). Absolute ages after Harland and others (1989). LNC, Lower Nicoya Complex; UNC, Upper Nicoya Complex; NA, neoautochthonous unit. Right column shows formation names and regions of occurrence. Asterisk: radiometric age (Barr and Escalante, 1969).

Cretaceous epoch, a neoautochthonous sequence of mainly calcareous sedimentary rocks containing pillow basalt and dolerite dikes formed (Fig. 2). Siliceous limestone of the Sabana Grande Formation, forming the basal part of the sedimentary sequence, are of late Campanian age (Sprechmann, 1984). Xenoliths of the older units are included in the dolerite dikes of the neoautochthonous sequence.

Geochemical Characteristics of the Basalts

The ophiolite complexes consist of a composite edifice of basaltic rocks with a wide range of chemical composition. Our genetic interpretation of the basalts is based on 110 chemical analyses and the results of Wildberg (1984). The geochemical data and sample locations are presented in Tables A and B of the GSA Data Repository.

Basalt of the Matapalo Nappe has a trace-element pattern similar to N-type MORB (normal-type mid-ocean ridge basalt; Pearce, 1982). It has a high field strength (HFS) element pattern (including Nb, P, Zr, Ti, Y) close to the unity line in the MORB normalized diagram (Fig. 5a and Wildberg, 1984). The enrichment in large ion lithophile (LIL) elements (Sr, K, Rb, Ba) relative to N-type MORB may be interpreted by hydrothermal alteration, submarine

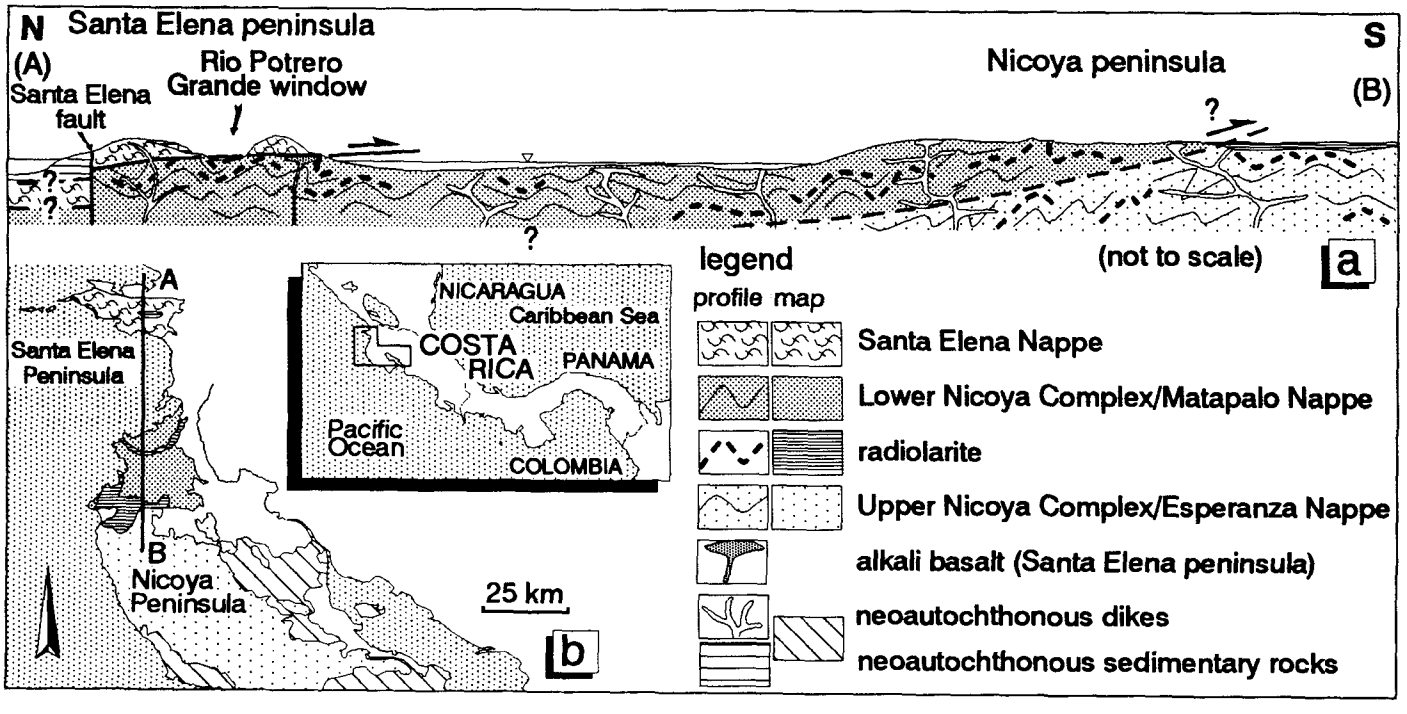


Figure 3. (a) Schematic cross section through the northern part of the Nicoya peninsula and the Santa Elena peninsula, Costa Rica. (b) Sketch map and location of cross section (Gursky and others, 1984).

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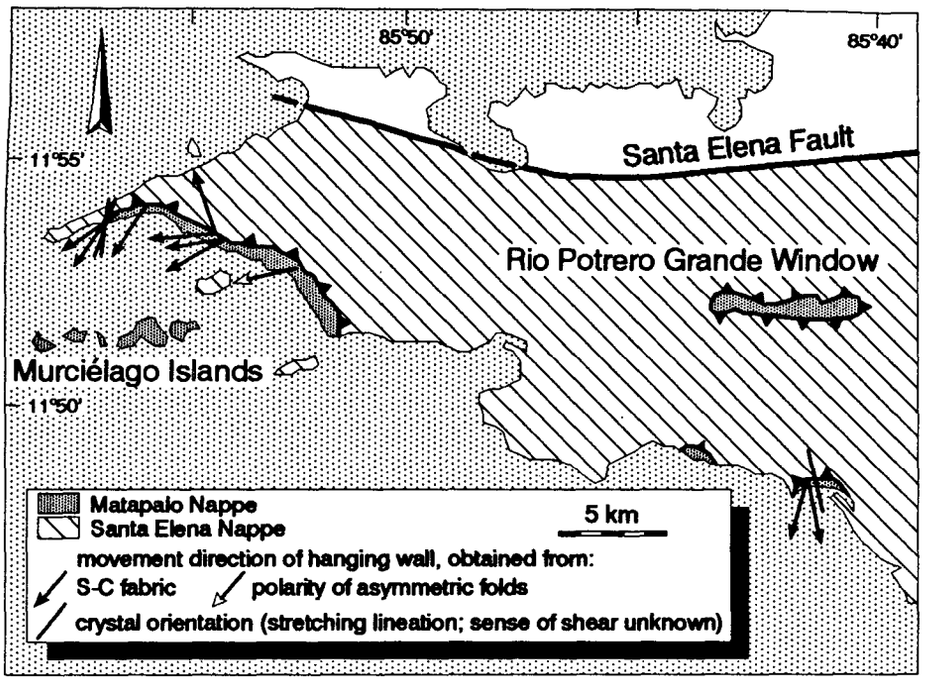


Figure 4. Geologic map of Santa Elena peninsula (modified after Tournon and Azéma, 1980), northwest Costa Rica, with structural data along the thrust of the Santa Elena Nappe (S-C fabric according to Berthé and others, 1979).

weathering, or, in connection with a slight depletion in HFS elements in most samples, as a weak subduction-related component. A subduction component would indicate that the mantle which generated the basaltic melt already contained these characteristics, inherited from preceding subduction activity along the western margin of the American continental crust (see Fig. 11 below). This is corroborated by the chondrite-normalized rare-earth element (REE) distribution, which corresponds to MORB patterns but has a slight increase in the light REE relative to N-type MORB (Fig. 5f; Henderson, 1984; Wilson, 1989). In the diagrams in Figure 6, these basalts are clearly distinct from the other groups. In the Ti-V diagram after Shervais (1982), they occupy a MORB position. As the basalts of the Matapalo Nappe are associated with radiolarian chert and plutonic rocks of the ophiolite sequence such as plagiogranite, we consider them to represent the oceanic basement and assign a mid-ocean ridge environment of formation to them. Basalts with an identical geochemical pattern also occur in the Esperanza Nappe, but are subordinate.

On the southern coast of the Santa Elena peninsula, pillow basalts rich in vesicles are associated with the Jurassic sedimentary cover of the oceanic basement. They have a within-plate alkaline trace-element pattern with strong enrichment in incompatible elements (Figs. 5b, 6; Pearce and others, 1981). The chondrite-normalized REE curve indicates significant enrichment in the light REE and major heavy to light REE fractionation (Fig. 5f). The abundant vesicles are evidence that the pillows formed in a maximum water depth of 1,000 m (Moore, 1975). Hyaloclastitic pillow block breccias indicate slopes of more than 15° inclination (Furnes and Fridleifsson, 1979). We interpret the alkali basalts as part of an intraplate seamount possibly associated with block faulting and uplift.

Basaltic rocks with a marked enrichment in LIL elements and a moderate but significant depletion in HFS elements relative to MORB (Fig. 5c) are widespread in the Esperanza Nappe (Upper Nicoya Complex of Wildberg and others, 1981) and occur as dikes and sills in all three units ("younger basalts"; Fig. 2). Such a pattern is known from primitive island-arc tholeiites (IAT) formed near the volcanic front (Fig. 5e; Pearce, 1983; Wilson, 1989) but also from other tholeiitic sequences. Judging from the HFS pattern, the mantle source was suboceanic, because subcontinental lithosphere would cause enrichment in P relative to MORB (Pearce, 1983). Calc-alkaline basalt formed behind the volcanic front would also be enriched in P and possibly in Zr (Fig. 5e; Pearce, 1983). Diagrams in Figure 6 display a distinct grouping from the MORB of the oceanic basement. The tholeiitic trend is confirmed by the lack of fractionation of Ti (Fig. 5c), due to late precipitation of magnetite in such melts. Negative correlation of the HFS elements with Cr is caused by slight differentiation of the melts and cannot be explained by various degrees of mantle melting. The REE patterns are depleted relative to the MORB of the basement (Fig. 5f). Island-arc tholeiites are typically depleted in the whole range of REE (Wilson, 1989). An island-arc affinity of these basalts is also corroborated by slightly elevated ^{87}Sr values relative to the MORB samples (Wildberg, 1984). Submarine weathering can be excluded as an explanation of increased LIL contents and ^{87}Sr values because not only effusive basalts but also dikes and sills contained in the basement and its sedimentary cover exhibit these element patterns.

The basaltic rocks with distinct LIL enrichment and HFS depletion (Fig. 5c) have similar chemical patterns as the MORB of the oceanic basement (Fig. 5a). They yielded, however, a Late Cretaceous radiometric age (Barr and Escalante, 1969; see above), penetrate as dikes and sills the oceanic basement and radiolarian chert sequence, and are therefore considerably younger than the basement MORB ("younger basalts" in Fig. 2). We interpret them as a distinct group of tholeiitic basalts containing an IAT component. We refer to them as basalt with IAT affinity.

Another group of basalt which is restricted to the Esperanza Nappe and occurs in association with the basalt with IAT affinity has the characteristics of within-plate tholeiites (WPT) or basalts which are transitional between MORB and WPT (Figs. 5d, 6). They are less enriched in LIL and HFS elements than are the alkali basalts described above and are clearly distinguishable from the basalt with IAT affinity by their enriched HFS pattern. The REE pattern reflects unfractionated heavy REE and low abundances of light REE. This indicates that they may derive from a similar shallow mantle source as the MORB of the oceanic basement did earlier but experienced fractionation of olivine, clinopyroxene, plagioclase, and magnetite (Wilson, 1989). The similar REE patterns of the MORB, IAT, and WPT type rocks suggest that these basalts are related to each other and have a more or less

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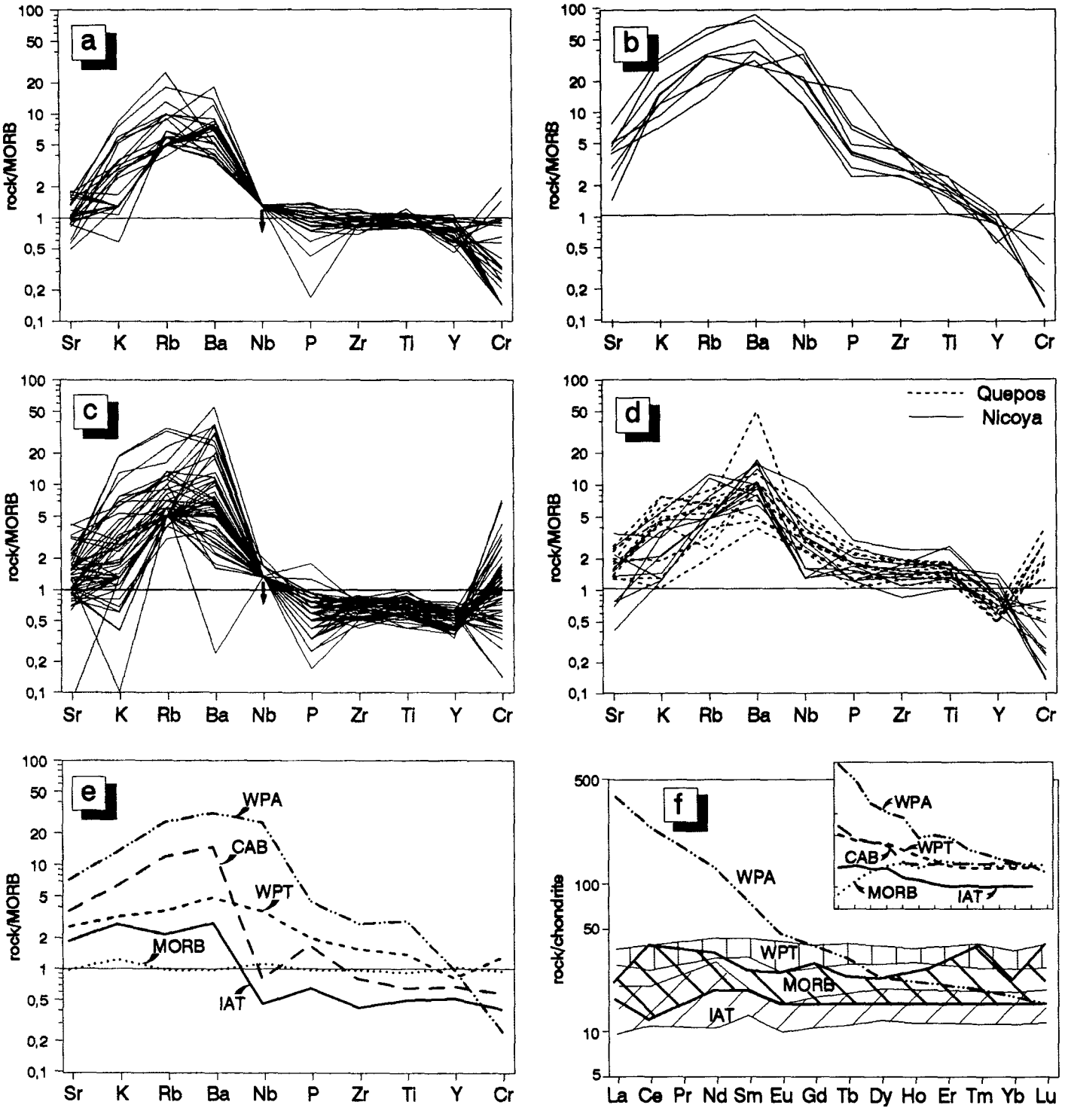


Figure 5. Geochemical results from basalts of the Central American ophiolite complexes (a–d) MORB normalized spiderdiagrams showing characteristics of mid-ocean ridge basalts (a), within plate alkali basalts (b), island-arc tholeiites (c), and within-plate tholeiites (d). Normalization values from Pearce (1982). Arrows indicate values beneath detection limit. (e) Typical trace-element patterns of basaltic rocks for different geotectonic settings (Pearce, 1982, 1983). (f) Chondrite normalized rare-earth-element data from selected samples from all four groups of basaltic rocks. Inset shows typical patterns for different geotectonic settings (Henderson, 1984; Wilson, 1989).

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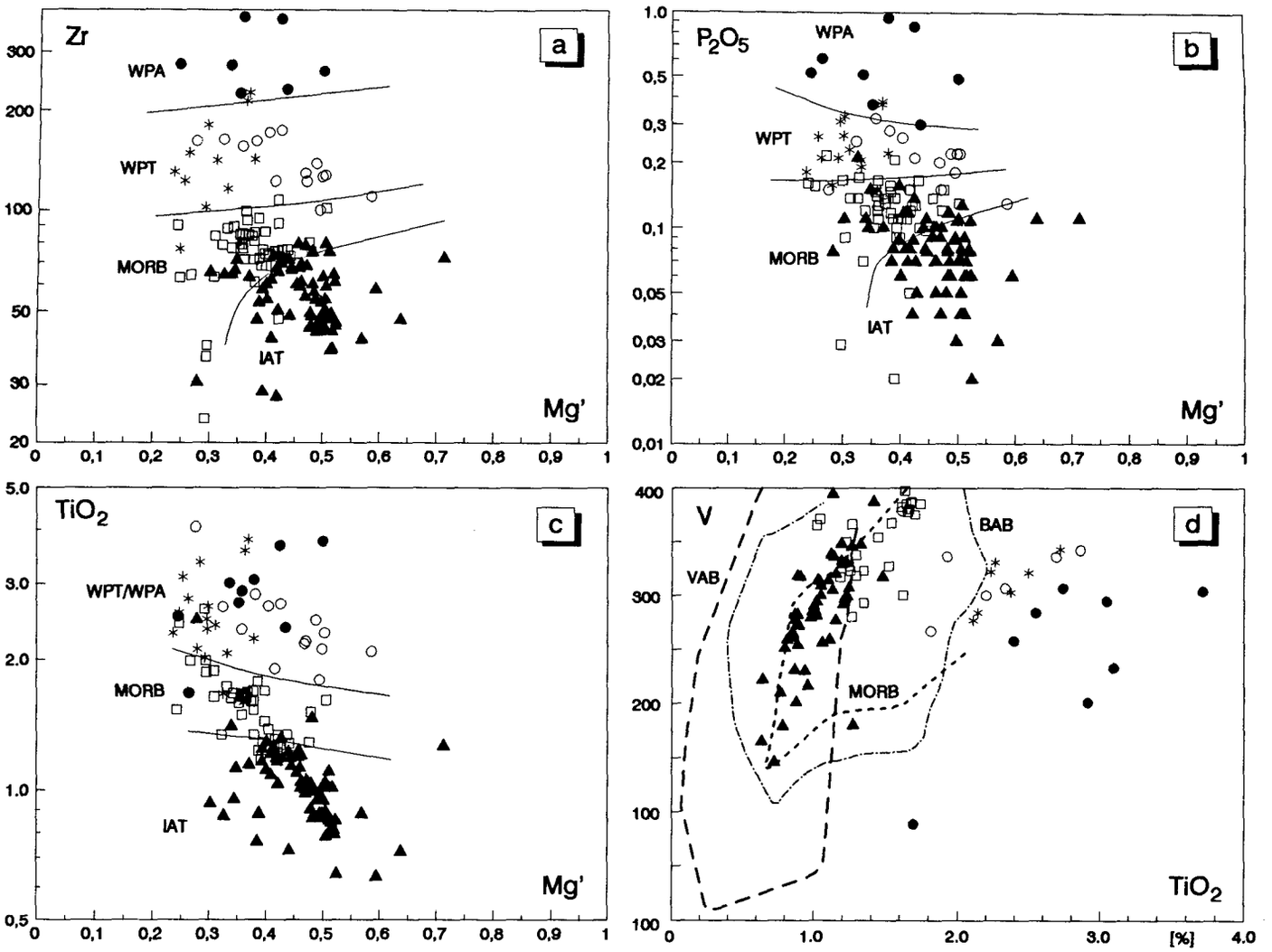


Figure 6. (a–c) Diagrams of HFS elements versus Mg number [Mg′ = MgO/(MgO + FeO_{tot}) where $Fe^{3+}/(Fe^{2+} + Fe^{3+})$ is assumed to equal 0.1; Basaltic Volcanism Study Project, 1981]. Open squares, MORB-type basalts; filled triangles, basalts with IAT affinity; asterisks, basalts with WPT affinity (Nicoya); open circles, WPT-type basalts from Quepos; filled circles, WPA-type basalts. (d) TiO_2-V diagram after Shervais (1982); MORB, VAB (volcanic arc basalt), and BAB (back-arc basin basalt) fields are shown. For explanation, see text.

common high-level source in probably depleted mantle which was metasomatically enriched by secondary processes.

We attribute the basalts with WPT and IAT affinity to the Caribbean "sill event" ("younger basalts" in Fig. 2). Basalts elsewhere in the Caribbean emplaced by the "sill event" between Late Neocomian and Santonian (Donnelly and others, 1973; Burke and others, 1978) have geochemical characteristics of both MORB and within-plate tholeiite (Bence and others, 1975; Donnelly, 1985). The within-plate tholeiite or transitional MORB/WPT of the Esperanza Nappe is found mainly in the central part of the Nicoya peninsula, whereas basalt with IAT affinity is exposed along the coast (Wildberg, 1984) in a position closer to the paleotrench. We therefore interpret the occurrence of basalt with IAT affinity to the influence of the Middle American subduction zone.

THE OPHIOLITE COMPLEXES SOUTH OF THE COSTA RICA FRACTURE ZONE

The Costa Rica Fracture Zone is a sinistral strike-slip fault (Fig. 1) active since middle or late Miocene time (Berrangé and Thorpe, 1988). The ophiolite complexes to the south (that is, the complexes of Quepos, Osa, Burica, Soná, and Azuero) consist mainly of basalt, dolerite, and sedimentary rocks. As in the Esperanza Nappe, the basalt and dolerite have predominantly IAT affinity. They contain xenoliths from the oceanic basement (MORB). K/Ar ages from Osa basalt and dolerite yielded ages of 78, 60, and 44 Ma (Berrangé and others, 1989). The sedimentary rocks are comparable in age and facies to the neoautochthonous strata north of the Costa Rica Fracture Zone. They range in age from Campanian to Paleocene (Baumgartner, 1987).

In addition, N-type MORB, oceanic basement on which the island arc evolved, and WPT (suite of Quepos; Figs. 1, 5d, 6) are also present. The pillows of the Quepos basalts are rich in vesicles, indicating moderate water depth. Strongly weathered reddish WPT that top pillow basalts have flow structures and lack chilled margins. They contain breccia layers with reworked basaltic fragments. MacPherson (1983) described similar phenomena in the Franciscan

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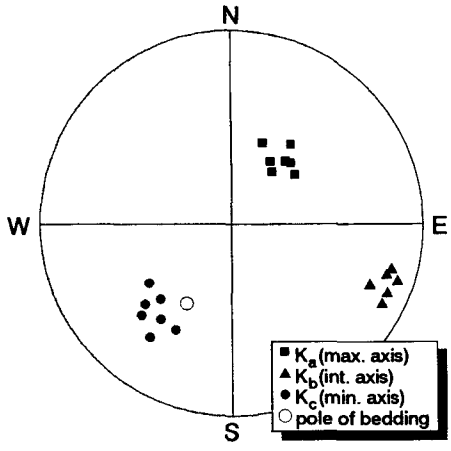


Figure 7. Principal axes of anisotropy of magnetic susceptibility (AMS) measured in a basalt flow from the Azuero peninsula (sample 103, see Table 1). Pole of bedding in adjacent sedimentary rocks coincides with short AMS (κ_C) axes.

Complex and explained them in terms of subaerial eruptions on a large seamount. We interpret the WPT suite of Quepos as belonging to an oceanic island. Azéma and others (1979) and Baumgartner and others (1984) reported a latest Cretaceous to early Paleocene age for intrapillow and overlying sedimentary rocks.

Individual rotations as evaluated by the paleomagnetic results subdivide the ophiolite complexes south of the Costa Rica Fracture Zone into three blocks: the Osa, Burica, and Azuero blocks.

PALEOMAGNETIC RESULTS

The results of 584 samples from 98 sites drilled in the ophiolite complexes and overlying strata along the Pacific coast of Costa Rica and western Panama were used for the paleomagnetic reconstruction (Tables 1, 2). An average of six oriented cores was taken from each site. Some sites (92) were discarded for one of several reasons: (1) α_{95} values of more than 30°; (2) instabilities during demagnetization; (3) viscous remanent magnetization.

Magnetization was measured with a Digico spinner magnetometer. Basaltic and ultramafic samples were demagnetized using alternating fields (AF). AF demagnetization was carried out with a Highmoor AF demagnetizer. A large number of pilot samples was demagnetized by applying increasing fields of 5 to 100 mT (steps were 5 mT from 5 to 20 mT and 10 mT from 20 to 100 mT). The sedimentary samples were thermally demagnetized in a Schonstedt oven using the following steps: 50, 100, 150, 200, 300, 350, 450, 500, 600, 650, and 670 °C. The characteristic remanent magnetization (ChRM) became generally stable in the range of 10 to 50 mT and 300 to 450 °C, respectively. A slight overprint, in most cases of the actual geomagnetic field, was removed by the applied demagnetization. The results were examined in Zijderveld diagrams.

Problems arose in several cases with the bedding correction of basalt samples. Statistically significant improvement of the site grouping for tilt-corrected characteristic remanent magnetization (ChRM) vectors was obtained by measuring the anisotropy of the magnetic susceptibility (AMS) and correlating the short axis of the obtained ellipsoid with the pole of the bedding plane (see MacDonald and Ellwood, 1987). A test example is shown in Figure 7. Positive fold tests indicate that the remanences are primary, which is corroborated by antiparallel normal and reversed magnetic vectors in each group of rocks.

The paleomagnetic results are summarized in Tables 1 and 2 and Figures 8 and 9. The tilt-corrected data reveal grouping of ChRM vectors for different tectonostratigraphic units (Fig. 8). We conclude that certain block rotations were carried out, although we do not fix exact figures

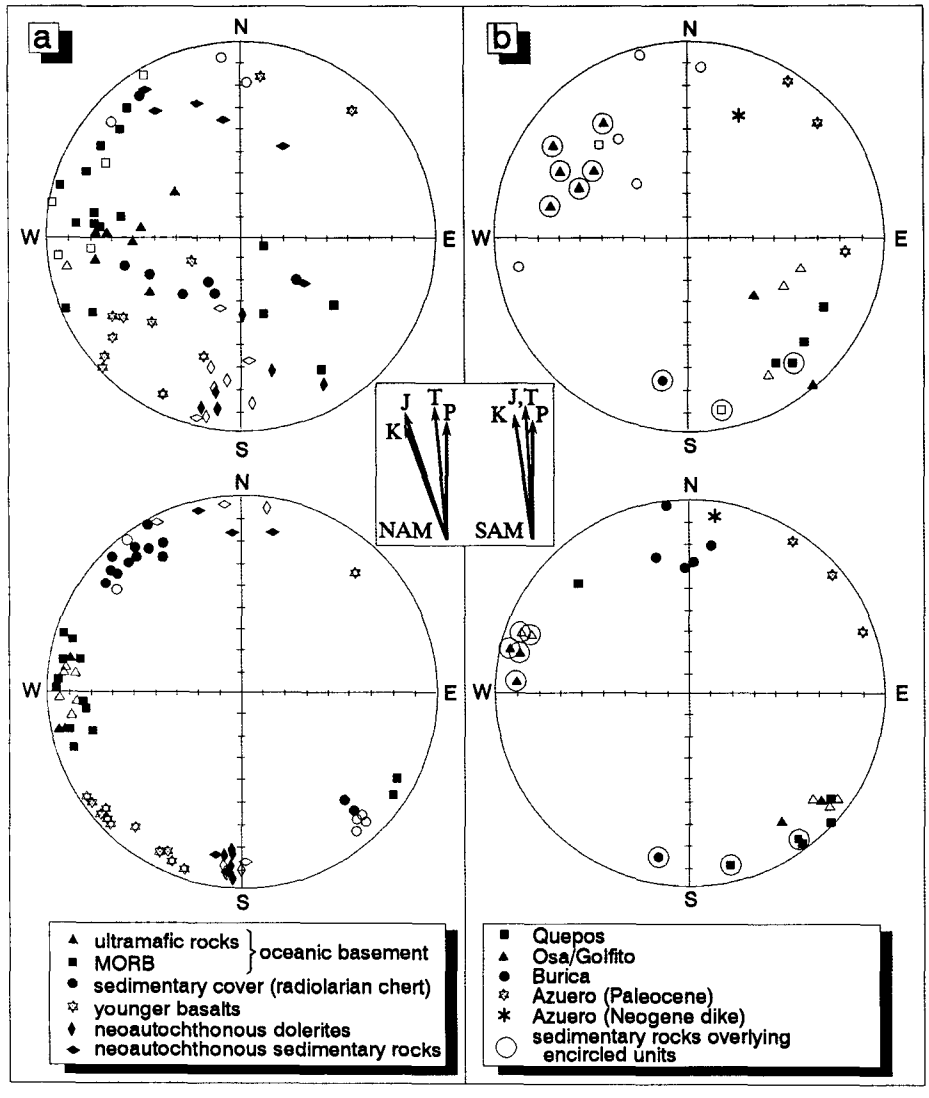


Figure 8. Uncorrected (upper row) and tilt-corrected (lower row) site mean directions of the ophiolite complexes of Central America, (a) north and (b) south of the Costa Rica Fracture Zone. Full symbols, positive inclination (lower-hemisphere projection). Open symbols, negative inclinations (upper-hemisphere projection). Inset: paleopole directions for adjacent North America (NAM) and South America (SAM) after Duncan and Hargraves (1984). J: Late Jurassic; K: Late Cretaceous; T: early Tertiary; P: present north direction.

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for the rotation angles, because of the scatter of site mean values.

Ophiolite Complexes North of the Costa Rica Fracture Zone

The mean directions of the ChRM of the oceanic basement (that is, the ultramafic rocks of the Santa Elena Nappe and the MORB of the Matapalo and Esperanza Nappes) are almost identical. The radiolarites have a somewhat different grouping (Fig. 8a). The low inclination values imply that the magnetic remanence was imprinted in a position near the equator. The present ChRM vectors point in a west to northwest direction and have several reversals. The declination data indicate rotation of the oceanic basement and its sedimentary cover. We assume counterclockwise rotation, because we consider it unlikely that the thinly bedded radiolarite sequence, only a few tens of meters thick, independently rotated by an angle of about 150°. If this assumption is correct, the sedimentary cover was less rotated (about 30° relative to the paleopole of South America) than the basement (about 80°) due to offscraping along the basement/cover interface (see above). The mean inclination values indicate a slight northward shift in the time span the oceanic basement and its cover formed, that is, from early Middle Jurassic to early Late Cretaceous time, but error bars overlap considerably (Fig. 10).

Although the base of the Santa Elena Nappe was in a thermal state that allowed ductile deformation of amphibolite (about 500 °C), the interior of the nappe shows no penetrative deformation and metamorphic overprint. Partial serpentinization may have occurred along faults and cracks in the vicinity of the spreading center by convecting sea water. The ultramafic body probably cooled, receiving its remanent magnetization in this early stage, and was not sufficiently heated in its interior during nappe emplacement to alter this early magnetization. An alternative interpretation would be that the remanence formed during cooling as a consequence of the nappe thrusting in Late Cretaceous time. We prefer the first interpretation, because the Santa Elena Nappe did not rotate after its emplacement as indicated by the neoautochthonous sedimentary rocks.

The prevailing Late Cretaceous basalts of the Esperanza Nappe (that is, the basalt with IAT and WPT chemical affinities) indicate slight clockwise rotation, although the reliability of these data is poor (Fig. 8a). If this clockwise rotation of about 30° is not an artifact, then it means that the counterclockwise rotation of the older rocks was accordingly larger. The mean inclination (Table 2) again signifies formation in

TABLE 1. PALEOMAGNETIC RESULTS OF SAMPLE SITES FROM THE OPHIOLITIC COMPLEXES IN COSTA RICA AND PANAMA

Site	Age	N	DEC	INC	α_{95}	κ	Treat	Rock	Location	Longitude	Latitude
1. Ultramafic rocks (Santa Elena)											
343	Jur.	6	265	-9	18.0	15.2	af 15	ultra	SEL Rio Murciélago	85°43'10"	10°54'35"
344	Jur.	6	272	-5	21.4	10	af 15	ultra	SEL Rio Murciélago	85°42'50"	10°54'30"
350	Jur.	8	257	12	5.1	120	af 15	ultra	SEL Rio Murciélago	85°42'40"	10°53'50"
351	Jur.	6	265	-13	11.0	52	af 15	ultra	SEL Rio Murciélago	85°42'40"	10°53'50"
352	Jur.	6	261	-9	7.7	76	af 15	ultra	SEL Rio Murciélago	85°42'35"	10°53'35"
385	Jur.	7	285	-11	4.9	150	af 15	ultra	SEL Pa. Morrito	85°52'25"	10°55'25"
386	Jur.	5	282	5	3.0	666	af 15	ultra	SEL Pa. Morrito	85°52'20"	10°55'25"
387	Jur.	5	282	3	6.8	127	af 15	ultra	SEL Pa. Morrito	85°52'10"	10°55'40"
2. Basaltic basement											
3	Jur.	11	264	-6	7.5	37	af 15	flow	NIC Pa. Ballena	85°32'00"	10°35'25"
14	Jur.	11	246	3	9.2	24.5	af 15	pill	NIC Pa. Penca	85°47'20"	10°29'00"
32	Jur.	7	296	-22	10.8	31.6	af 15	pill	NIC Tambor	85°01'00"	09°43'25"
40	Jur.	11	276	16	3.6	163	af 15	pill	HER Pa. Guapinol	84°37'25"	09°35'00"
44	Jur.	7	278	-18	5.8	107	af 15	flow	HER Pa. Guapinol	84°37'35"	09°35'20"
45	Jur.	5	267	-3	3.2	559	af 15	flow	HER Pa. Guapinol	84°37'20"	09°35'35"
131	Jur.	2	254	18	17.6	204.1	af 15	pill	NIC Pa. Camaronal	85°25'55"	09°51'20"
146	Jur.	6	285	9	3.0	509.5	af 15	sill	SEL Isla San José	85°54'45"	10°51'50"
147	Jur.	6	300	8	11.0	125.6	af 15	pill	SEL Isla San José	85°54'40"	10°51'45"
148	Jur.	4	288	2	8.3	132.2	af 15	pill	SEL Isla San José	85°54'45"	10°51'40"
324	Jur.	5	119	12	16.0	17.8	af 15	flow	NIC Pa. Carbon	85°52'10"	10°21'00"
326	Jur.	6	127	-7	17.9	14.2	af 15	flow	NIC Pa. Carbon	85°52'00"	10°21'00"
361	Jur.	6	260	15	19.2	12.3	af 15	pill	NIC Pa. Mostrenco	85°41'05"	10°36'50"
3. Radiolarites and intercalated basalts (Late Jurassic–Early Cretaceous)											
69	Tithon.	8	330	17	19.6	12.7	af 15	pill	NIC Pla. Lagartillo	85°51'00"	10°13'30"
73	Tithon.	4	306	14	5.4	285.5	af 15	pill	NIC Pla. Lagartillo	85°50'55"	10°13'35"
201	Ea. Cret.	7	136	27	9.1	44.2	td 350	radi	NIC Pa. Sabanna	85°49'45"	10°24'55"
270	Ea. Cret.	7	141	21	6.4	89	td 350	radi	NIC Sardinal	85°39'50"	10°32'05"
280	Ea. Cret.	8	302	-20	17.6	10.1	af 150	pill	NIC Pa. Cacique	85°41'55"	10°34'15"
283	Ea. Cret.	7	141	-9	4.2	211.1	af 150	flow	NIC Sardinal	85°39'50"	10°32'05"
284	Ea. Cret.	6	140	-14	10.9	38	td 350	radi	NIC Sardinal	85°39'50"	10°32'05"
310	Ea. Cret.	2	337	12	22.1	67.2	td 350	radi	NIC Nancital	85°38'05"	10°33'45"
327	Late Jur.	6	147	-9	17.9	14.2	af 15	flow	NIC Pa. Carbon	85°52'10"	10°21'00"
801	Ea. Cret.	8	141	-15	9.3	35.5	td 450	radi	NIC Pa. M. del Barco	85°33'15"	10°37'10"
4. Radiolarites and intercalated basalts (Albian)											
139	Alb.	4	332	14	6.9	175	td 350	radi	SEL Pla. Carizal	85°54'35"	10°53'55"
140	Alb.	6	313	13	8.4	63.6	af 15	sill	SEL Pla. Carizal	85°54'25"	10°53'50"
141	Alb.	5	341	0	9.4	66.9	td 350	radi	SEL Pla. Carizal	85°54'25"	10°53'50"
142	Alb.	4	328	-3	11.9	60.8	af 150	sill	SEL Pla. Carizal	85°54'25"	10°53'50"
5. Radiolarites and intercalated basalts (Late Cretaceous)											
49	Presant.	5	317	6	15.7	24	td 350	radi	HER Jacó	84°39'00"	09°37'15"
207	Santon.	9	304	17	3.5	216.6	td 400	radi	NIC Pa. Gorda	85°47'15"	10°32'10"
208	Santon.	10	342	12	8.4	33	td 350	radi	NIC Pa. Gorda	85°47'15"	10°32'10"
210	Santon.	18	316	16	7.0	28.3	af 15	flow	NIC Pa. Gorda	85°47'25"	10°32'10"
6. Younger basalts (Late Cretaceous)											
24	Presant.	5	225	-4	21.2	10.3	af 15	pill	NIC Tambor	85°00'25"	09°44'00"
28	Presant.	4	195	-4	22.5	9.2	af 15	pill	NIC Tambor	85°04'35"	09°38'55"
37	Presant.	6	206	-9	3.3	402	af 15	pill	HER Pla. Herradura	84°40'25"	09°38'55"
76	Presant.	6	229	-11	5.4	155	af 15	pill	NIC Pla. Herradura	85°48'00"	09°38'55"
79	Presant.	8	226	-2	7.2	62.1	af 15	pill	NIC Pla. Islita	85°24'55"	09°51'05"
80	Presant.	4	234	2	9.0	32.2	af 15	pill	NIC Brasilito	85°47'50"	10°25'40"
123	Presant.	8	208	12	11.2	24.3	af 15	pill	NIC Brasilito	85°47'50"	10°25'40"
144	Presant.	10	44	-21	21.0	5.6	af 15	dike	SEL Isla San José	85°54'40"	10°51'45"
329	Presant.	2	204	6	25.0	99.2	af 15	pill	NIC Ostional	85°43'15"	10°00'45"
331	Presant.	7	219	-16	6.4	87.7	af 15	pill	NIC Pa. Agujas	85°44'50"	10°02'00"
346	Presant.	7	224	-2	8.3	8.3	af 15	dike	SEL Rio Murciélago	85°42'50"	10°54'30"
362	Presant.	8	228	3	21.0	7.1	af 15	flow	NIC Pa. Mostrenco	85°41'05"	10°36'50"
7. Neoautochthonous basalts (Late Cretaceous)											
38	Sant./Camp.	9	187	16	11.7	19.3	af 15	flow	HER Pla. Herradura	84°40'20"	09°39'00"
39	Sant./Camp.	6	193	-12	9.5	50.5	af 15	flow	HER Pla. Herradura	84°40'15"	09°39'00"
125	Sant./Camp.	6	187	18	4.3	247.3	af 15	flow	NIC Pla. Islita	85°20'55"	10°02'05"
190	Sant./Camp.	7	177	16	6.5	86.1	af 15	flow	NIC Brasilito	85°47'40"	10°25'40"
191	Ea. Cret.	6	187	16	16.3	17	td 350	radi	NIC Brasilito	85°47'40"	10°25'40"
291	Sant./Camp.	7	186	5	22.3	8.2	af 15	flow	NIC Pa. Salinas	85°47'45"	10°26'50"
292	Sant./Camp.	6	191	-16	14.1	22.8	af 15	flow	NIC Pa. Salinas	85°47'45"	10°26'50"
295	Ea. Cret.	5	12	-3	3.5	471	td 350	radi	NIC Pa. Conchal	85°49'15"	10°24'45"
800	Sant./Camp.	7	183	18	11.8	26.3	af 15	dike	NIC Pa. M. del Barco	85°33'15"	10°37'10"
3700	Sant./Camp.	4	192	-12	3.3	402.1	af 15	pill	HER Pla. Herradura	84°40'25"	09°38'55"

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TABLE 1. (*Continued*)

Site	Age	N	DEC	INC	α_{95}	κ	Treat	Rock	Location	Longitude	Latitude
8. Neoautochthonous sedimentary rocks (Late Cretaceous)											
2	Camp.	6	186	8	18.1	13.9	td 400	calc	NIC Pa. Gigante	84°55′00″	09°54′30″
119	Camp.	3	332	−1	19.6	44.9	td 350	calc	NIC Pa. Islita	85°29′00″	09°51′50″
301	Camp.	7	354	−1	32.0	4.2	td 350	calc	NIC Bevedero	85°12′55″	10°22′25″
353	Camp.	5	179	−11	11.7	42.8	td 350	calc	NIC Ba. Murciélago	84°56′35″	09°45′15″
354	Camp.	7	16	20	5.0	149.9	td 350	calc	NIC Ba. Murciélago	84°56′35″	09°45′15″
355	Camp.	4	348	8	4.2	514	td 350	calc	NIC Ba. Murciélago	84°56′20″	09°45′20″
356	Camp.	6	359	15	6.3	112.2	td 350	calc	NIC Ba. Murciélago	84°56′20″	09°45′20″
358	Camp.	5	197	18	28.0	7.6	td 350	calc	NIC Ba. Murciélago	84°56′15″	09°45′20″
373	Maestr.	5	189	15	21.6	12.7	td 350	calc	BUR Pa. Banco	83°08′20″	08°23′55″
9. Quepos (Late Cretaceous)											
82	Maestr.	5	170	10	27.0	8.2	td 350	calc	QUE Pa. Quepos	84°10′15″	09°23′25″
83	Maestr.	6	149	11	27.3	8.5	td 350	calc	QUE Pa. Quepos	84°10′15″	09°23′25″
84	Maestr.	6	316	−12	8.5	62.1	af 15	flow	QUE Pla. Espadilla	84°10′20″	09°23′20″
89	Maestr.	4	131	−1	10.2	23	af 15	pill	QUE Pa. Quepos	84°10′15″	09°23′20″
90	Maestr.	7	121	13	15.0	40.1	af 15	pill	QUE Pa. Quepos	84°10′15″	09°23′20″
312	Maestr.	6	149	−8	7.7	61.5	af 15	flow	QUE Pla. Man. Ant.	84°10′30″	09°22′25″
10. Osa/Golfito (Late Cretaceous)											
159	Precamp.	7	124	−9	8.1	57.2	af 15	pill	OSA Pa. Ganadito	83°38′55″	08°44′05″
160	Precamp.	6	128	−17	3.7	328.0	af 15	pill	OSA Pa. Ganadito	83°39′00″	08°44′00″
161	Precamp.	5	153	19	7.8	96.1	td 350	radi	OSA Pa. Agujitas	83°40′45″	08°41′50″
162	Precamp.	8	127	12	19.2	9.3	af 15	flow	OSA Pa. Agujitas	83°40′40″	08°41′55″
168	Precamp.	7	134	−9	15.8	15.5	af 15	pill	GOL Pa. Curupacha	83°13′10″	08°37′55″
11. Golfito, sedimentary rocks (Late Cretaceous)											
173	Camp.	3	282	6	22.7	31.5	td 350	calc	GOL Pa. Agua Fria	83°12′40″	08°37′55″
174	Camp.	8	288	−6	12.4	20.1	td 350	calc	GOL Pa. Agua Fria	83°12′40″	08°37′55″
175	Camp.	3	288	−8	17.4	50.5	td 350	calc	GOL Pa. Agua Fria	83°12′35″	08°37′55″
176	Camp.	3	323	−2	16.8	54	td 350	calc	GOL Pa. Agua Fria	83°12′35″	08°37′55″
177	Camp.	6	272	10	26.4	7.7	td 400	calc	GOL Pa. Agua Fria	83°12′35″	08°37′55″
178	Camp.	3	282	2	22.7	6.9	td 400	calc	GOL Pa. Agua Fria	83°12′35″	08°37′55″
12. Burica (Paleocene)											
367	Paleoc.	7	358	29	18.3	11	af 15	pill	BUR Pa. Banco	83°07′30″	08°20′35″
368	Paleoc.	6	337	27	7.1	90.5	af 15	pill	BUR Pa. Banco	83°07′30″	08°20′35″
369	Paleoc.	8	1	27	14.5	14.7	af 15	pill	BUR Pa. Banco	83°07′30″	08°20′35″
371	Paleoc.	8	17	−21	3.5	249.2	af 15	pill	BUR Pa. Banco	83°07′45″	08°20′50″
372	Paleoc.	7	352	0	17.8	11.6	af 15	pill	BUR Pa. Banco	83°07′45″	08°20′50″
13. Azuero (Paleocene - Miocene)											
96	Mioc.	5	13	2	5.2	218.1	af 15	dike	AZU Pla. Pedasi	80°40′–	07°15′–
102	Paleoc.	7	95	−5	14.1	19.3	af 15	flow	AZU Tonosi	80°31′–	07°14′–
103	Paleoc.	5	32	3	12.7	36.3	af 15	flow	AZU Tonosi	80°31′–	07°14′–
104	Paleoc.	3	68	4	13.6	82.1	af 15	flow	AZU Tonosi	80°31′–	07°14′–

Note: N, number of samples. DEC, declination. INC, inclination. α_{95}, 95% confidence circle. κ, precision parameter. Treat, treatment of demagnetization: afd 15, alternating field demagnetization with 15 mT; td 350, thermal demagnetization with 350 °C. Rocks: amph, amphibolite; calc, limestone; dike, dolerite dike; flow, massive basalt flow; pill, pillow basalt; radi, radiolarite; sill, basaltic sill; ultra, ultramafic rock. Locations: SEL, Santa Elena peninsula; NIC, Nicoya peninsula; HER, Herradura; QUE, Quepos; OSA, Osa peninsula; GOL, Golfito; BUR, Burica peninsula; AZU, Azuero peninsula; Ba., Bahia; Pa., Punta; Pla., Playa. Latitude and longitude, coordinates of sites.

TABLE 2. MEAN VALUES OF PALEOMAGNETIC RESULTS OF THE DIFFERENT TECTONIC UNITS

Unit	DEC	INC	α_{95}	κ	Number of samples	Number of sites
1. Santa Elena Nappe	271	−4	12.4	20.2	49	8
2. Basaltic basement	274	2	10.5	15.6	87	13
3. Rad./basalts (Late Jur.-Early Cret.)	320	0	16.5	13.5	63	10
4. Rad./basalts (Albian)	329	6	17.0	29.5	24	5
5. Rad./basalts (Late Cretaceous)	320	9	18.7	24.3	37	3
6. Younger basalts (Late Cretaceous)	217	−4	8.8	24.2	75	12
7. Neoautochth. basalts (Late Cret.)	187	−3	8.5	32.4	52	10
8. Neoautochth. sed. rocks (Late Cret.)	359	8	11.9	16.5	48	9
9. Quepos	142	2	16.9	21.4	34	6
10. Osa/Golfito	133	0	18.8	16.7	33	5
11. Golfito sediments	289	1	13.8	23.7	26	6
12. Burica	357	12	19.6	15.4	36	5
13. Azuero	52	1	23.4	15.7	20	4
Overall			15.1	20.7	584	96

Note: for explanations, see Table 1.

a position near the equator. All sites but one have reversed magnetic polarity, assuming the minimum rotation angle. We interpret this in terms of age brackets for these basalts and dolerites between the Cretaceous Quiet Zone with predominantly normal magnetic polarity (Aptian to Santonian, 123–83 Ma; Harland and others, 1989) and the overlying strata with Early Campanian age (Sabana Grande Formation: Schmidt-Effing, 1979; Sprechmann, 1984). Therefore formation of these rocks in the 33r reversed polarity chron (83–79 Ma) that followed the Cretaceous Normal Polarity Superchron is inferred, which is consistent with radiometric dating (see above).

The neoautochthonous sedimentary and volcanic rocks of Campanian to Paleocene age do not show any significant rotation relative to the North American or South American paleopole. An equatorial place of formation is again indicated by the inclination values. Both normal and reversed magnetic polarities occur. The results are consistent with those of Gose (1983).

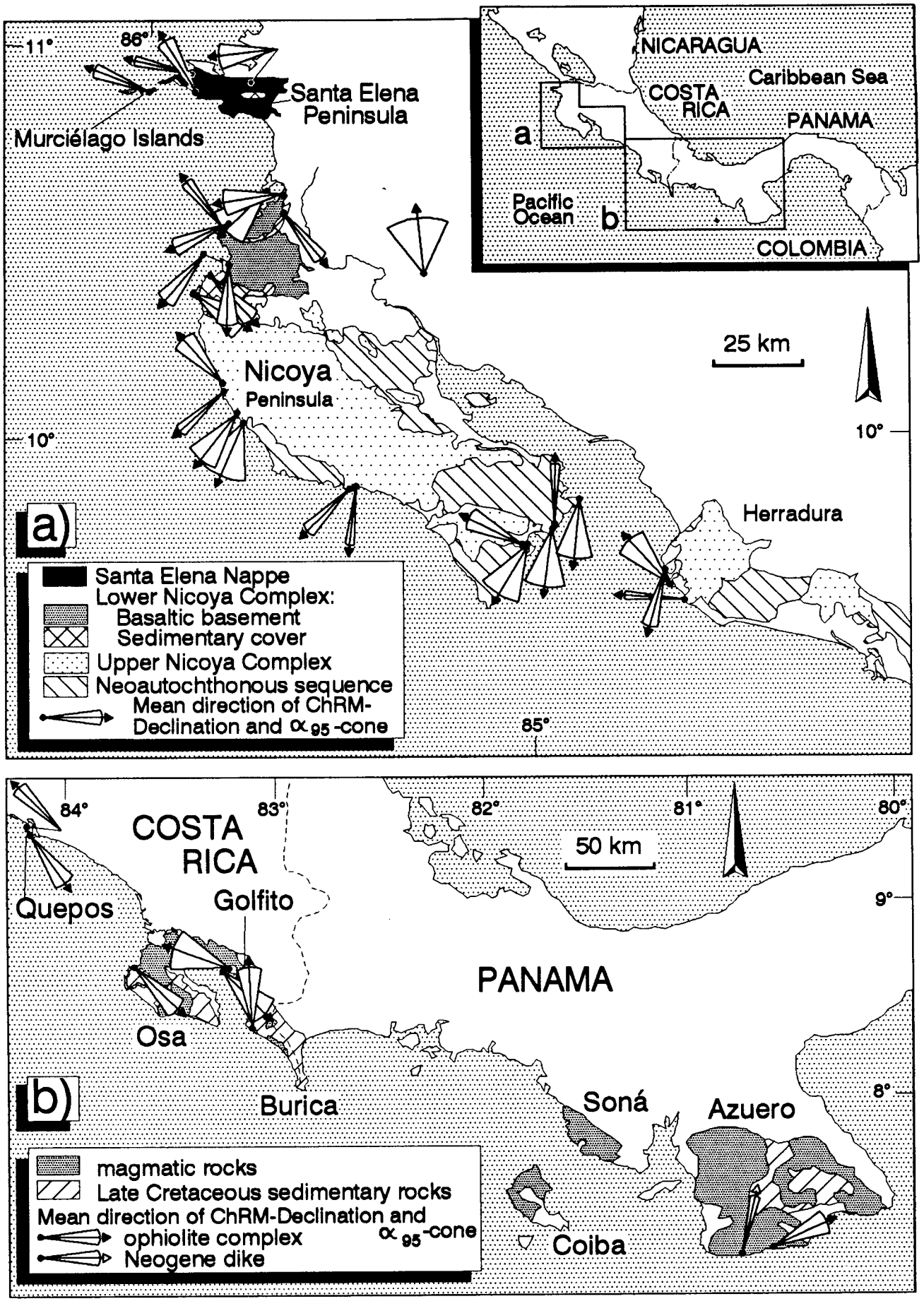
Ophiolite Complexes South of the Costa Rica Fracture Zone

The paleomagnetic data of the ophiolite complexes south of the Costa Rica Fracture Zone differ from those north of the fracture zone in that they reveal Tertiary rotation of individual blocks. No rocks could be sampled which unequivocally belong to the oceanic basement and its cover, and so we have no information about the pre–Late Cretaceous evolution.

The paleomagnetic data of basaltic and sedimentary rocks from Quepos and Osa show counterclockwise rotation, an equatorial place of formation, and both normal and reversed magnetic polarities (Fig.8b). The Campanian sedimentary rocks (Calizas de Golfito Formation; Sprechmann, 1984) north of Golfito suggest a slightly stronger rotation than the underlying basalt with island-arc affinity of the Osa complex. This is explained by offscraping and individual rotation of the sedimentary cover during tectonic processes. Inland from Golfito, we observed south-directed imbrication within the sedimentary cover (see Fig. 1). In analogy to the corresponding rocks of the Nicoya peninsula, the reversed magnetic polarity of the Osa basalts with IAT affinity suggest formation during the 33r polarity chron. The youngest rocks showing the counterclockwise rotation are the basalts and sedimentary rocks of Quepos (Paleocene to Eocene; Baumgartner and others, 1984).

The Burica block remained unrotated until the present time. The basaltic rocks formed in a Late Cretaceous period of normal magnetic po-

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Figure 9. Characteristic remanent magnetization declinations of representative sample localities in Central American ophiolite complexes (a) north and (b) south of the Costa Rica Fracture Zone (note different scales). Geology after Gursky and others (1984) and Case and others (1984).

larity. A sample of the overlying sedimentary sequence (Rivas Formation, Maestrichtian; Sprechmann, 1984; Corrigan, 1986) has reversed magnetic polarity.

The evolution of the Azuero Block is only poorly constrained by the paleomagnetic data. Three pre-Miocene basaltic rocks suggest, despite their scatter, clockwise rotation. A dike cutting Neogene breccia shows no significant deviation from the north direction, inferring that the rotation occurred prior to the emplacement of the dike. The inclination values again reveal an equatorial position during formation in periods of normal magnetic polarity.

PLATE-TECTONIC EVOLUTION OF CENTRAL AMERICA

The ophiolite complexes along the Pacific coast of Costa Rica and western Panama are considered to be the basement of the actual island arc that forms the Central American landbridge (Dengo, 1962; Weyl, 1980). Paleomagnetic results indicate that the ophiolite complexes (including the younger basalts that we attribute to the "sill event") and the overlying Late Cretaceous to early Tertiary sedimentary rocks formed in an equatorial position (Fig. 10). The mean inclination values reveal only minor latitudinal change since Jurassic time. Considering the error bars (Fig. 10), the latitudinal changes are not significant. The mean paleolatitudinal data assuming the lesser rotation angle for each unit (see above), however, imply a northward change over about 10° from the equator to the present position since the formation of the oceanic basement. This pattern approximately correlates with that of South America but is fundamentally different from that of the Pacific ocean floor (Farallon or Phoenix plates; Morgan, 1983; Duncan and Hargraves, 1984; MacDonald, 1990; Fig. 10). We interpret our paleomagnetic data in terms of formation of the Central American ophiolites in a position between the North and South American continental blocks, as opposed to formation as part of the Farallon/Phoenix plates.

An argument in favor of an inter-American origin of the oceanic basement of the ophiolite complexes is the turbiditic chert breccia within the radiolarian chert sequence of the oceanic sedimentary layer. The breccia age is Early or Middle Jurassic (DeWever and others, 1985; Baumgartner, 1987). Chert breccias are reported from escarpments in mid-ocean–ridge environments. Their presence is likely in the vicinity of a slowly spreading mid-ocean ridge with a rugged morphology and abundant fault scarps (Arcyana, 1975). Such an environment is expected in the Caribbean spreading system. Its formation is related to the formation of the central Atlantic Ocean in the Jurassic (Le Pichon and Fox, 1971), at which time the North and South American continental blocks started to separate (Van der Voo and French, 1974; Pindell and Dewey, 1982; Sclater and others, 1977; Burke and others, 1984; Pindell and others, 1988). Chert breccias are unlikely to occur in a fast-spreading mid-ocean–ridge environment with a smooth morphology, as is suggested for the spreading centers along the edges of the Farallon and Phoenix plates (Duncan and Hargraves, 1984).

Duncan and Hargraves (1984) and Mattson (1984) proposed that the Early to Middle Jurassic ocean floor preserved in the Central American ophiolite complexes came in from the Pacific region, overstepping the Central American subduction zone or pushing it eastward. Considering a Pacific origin of the Central American ophiolite complexes, assuming a steady northeastward drift of the Farallon/Phoenix plates (Mattson, 1984), and taking into account the arrival of the complexes in an equatorial position in Late Cretaceous time, the position of the nascent oceanic floor would be in the Southern Hemisphere at a latitude of more than 30°, according to the drift rates calculated by Duncan and Hargraves (1984) from hot-spot activity. Pindell and others (1988) assumed drift of the Phoenix plate relative to hot spots toward the south-southeast up to the Barremian stage. Ocean floor that formed in an equatorial position in Middle Jurassic time would drift to a southern latitude of more than 15° and again arrive at the equator in Late Cretaceous time. The paleolatitudes deduced from the paleomagnetic inclinations of the Central American ophi-

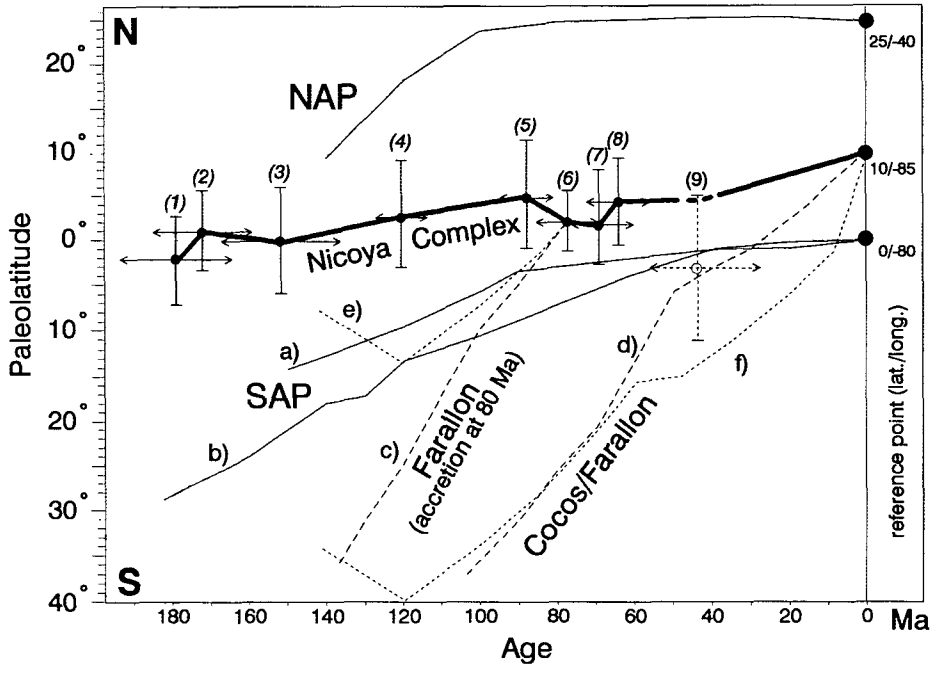


Figure 10. Paleolatitudes of ophiolite complexes in northwest Costa Rica against time. (1) Santa Elena ultramafic rocks; (2) basaltic basement (MORB); (3, 4, and 5) radiolarites and intercalated basalts; (6) younger basalts; (7) neoautochthonous basaltic dikes; (8) neoautochthonous sedimentary rocks; (9) neoautochthonous sedimentary rocks (Gose, 1983), age range not known. (1 through 8) These refer to same numbers in Tables 1 and 2. Reference point (10°N, 85°W) is on the Nicoya peninsula. Vertical lines are error bars (modified α_{95} values after Demarest, 1983); horizontal arrows show ranges in age. NAP, SAP, latitudinal paths of arbitrary reference points on the North and South American plates, respectively. NAP, SAP (a) after Duncan and Hargraves (1984); SAP (b) after Morgan (1983). Latitudinal paths of Farallon/Cocos plate after Duncan and Hargraves (1984) for accretion at 80 Ma (c) and 0 Ma (d), and after Pindell and others (1988) for accretion at 80 Ma (e) and 0 Ma (f).

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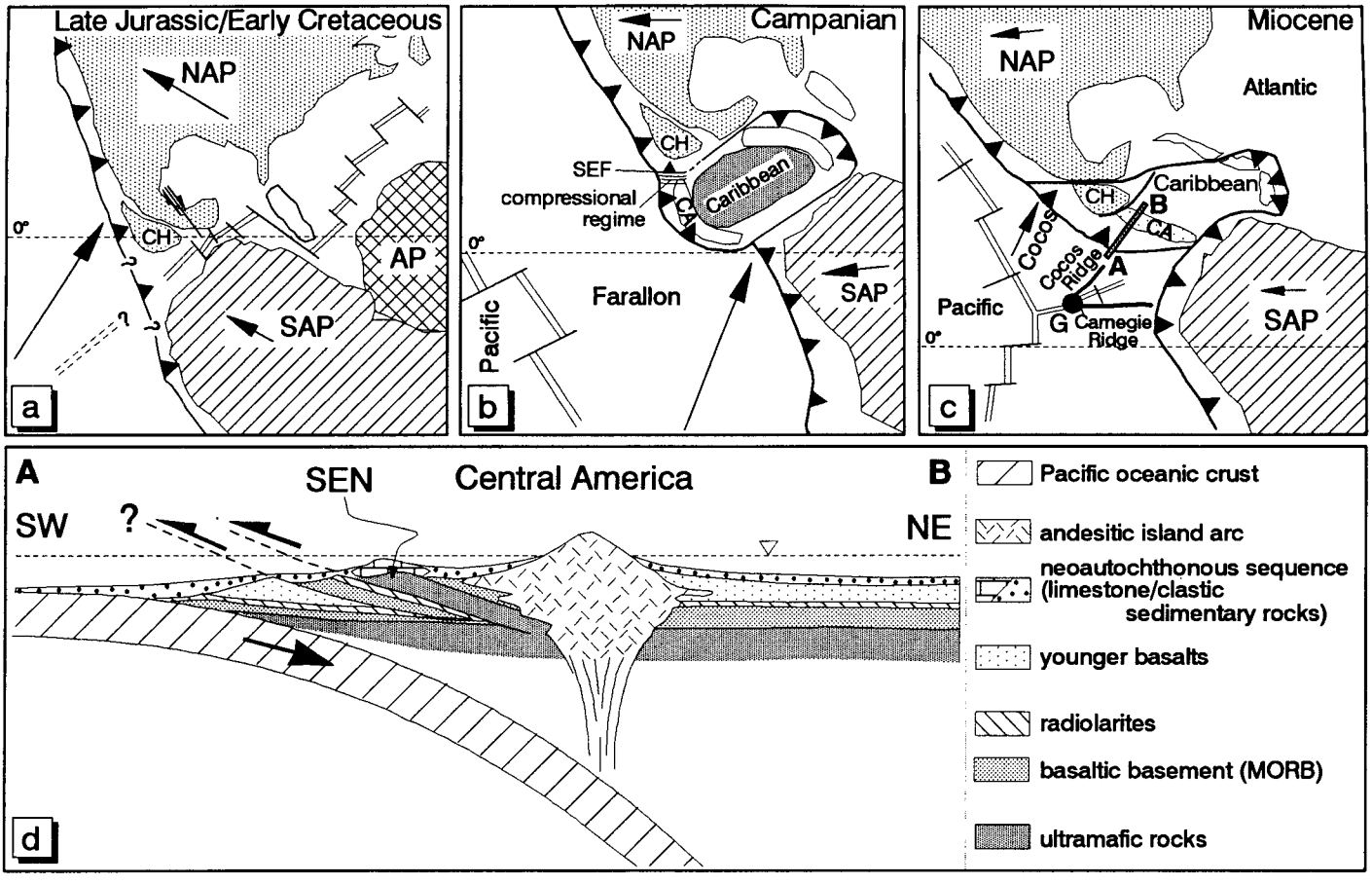


Figure 11. Plate-tectonic model for Central America (a–c) and cross section (d; early Tertiary; for location see c). Continents and Farallon plate after Duncan and Hargraves (1984), Chortis block after Anderson and Schmidt (1983), Nazca and Cocos plates after Malfait and Dinkelman (1972). AP, African plate. CA, Central America. CH, Chortis Block. G, Galapagos hot spot. NAP, North American plate. SAP, South American plate. SEF, Santa Elena fault. SEN, Santa Elena Nappe.

olites, however, do not support either solution (Fig. 10). We therefore propose a plate-tectonic model in which the ophiolite complexes take an inter-American position throughout their evolution (Fig. 11).

During the "sill event" that took place in the time interval between ca. 130 and 80 Ma (Donnelly, 1985), the Caribbean region was under extension, because the North American and South American plates were still moving apart at low drift rates (Pindell and others, 1988). This intraplate magmatic event spread over a large area, causing underplating of the oceanic crust and intrusion of large volumes of basaltic melt to form an oceanic plateau. The position of the Central American ophiolite complexes is marginal to the Caribbean plateau. Here the Central American subduction zone provided a source of island-arc magmatism (Fig. 11b). The magmatic accretion during the sill event is therefore controlled by both intraplate and subduction-related magma generation. The Late Cretaceous magmatism in the ophiolite complexes reflects this situation. The magmas raised from the same mantle region from which the oceanic basement derived earlier. This explains some common features of the basalts with different chemical fingerprints, as discussed above.

Much of the younger basalts is likely to have been emplaced in the Early Campanian (see above). The paleolatitude of these rocks agrees with a relative position to both South and North America similar to the present one. In the Campanian stage, Central America for the first time emerged substantially above the sea level. Terrestrial animal species were able to migrate between North and South America (Estes and Báez, 1985).

The tectonic event that is responsible for thrusting, folding, and block rotation occurred in Santonian and Campanian times. We interpret this as a response to the change of movement of South America relative to North America (Duncan and Hargraves, 1984). The ultramafic rocks of the Santa Elena Nappe are topped by late Campanian rudist limestone (Seyfried and Sprechmann, 1986). Neoautochthonous shallow-water sediments of Campanian age unconformably overlie the folded basalts and radiolarites of the Nicoya complex.

The north-northeastward drift of the Cocos plate caused the inactive Cocos Ridge to approach the Central American subduction zone in the Miocene epoch (Fig. 11c). We interpret the fragmentation and block rotation south of the Costa Rica Fracture Zone by the collision of the ridge with the Central American landbridge (Fig. 12; "Vogt hypothesis": Vogt, 1973; Vogt and others, 1976; McCabe, 1984). The collision of the Coiba and Cocos segments deformed and fragmented the upper-plate margin. Although the Burica block in front of the colliding ridge remained unrotated, the Azuero and Osa blocks

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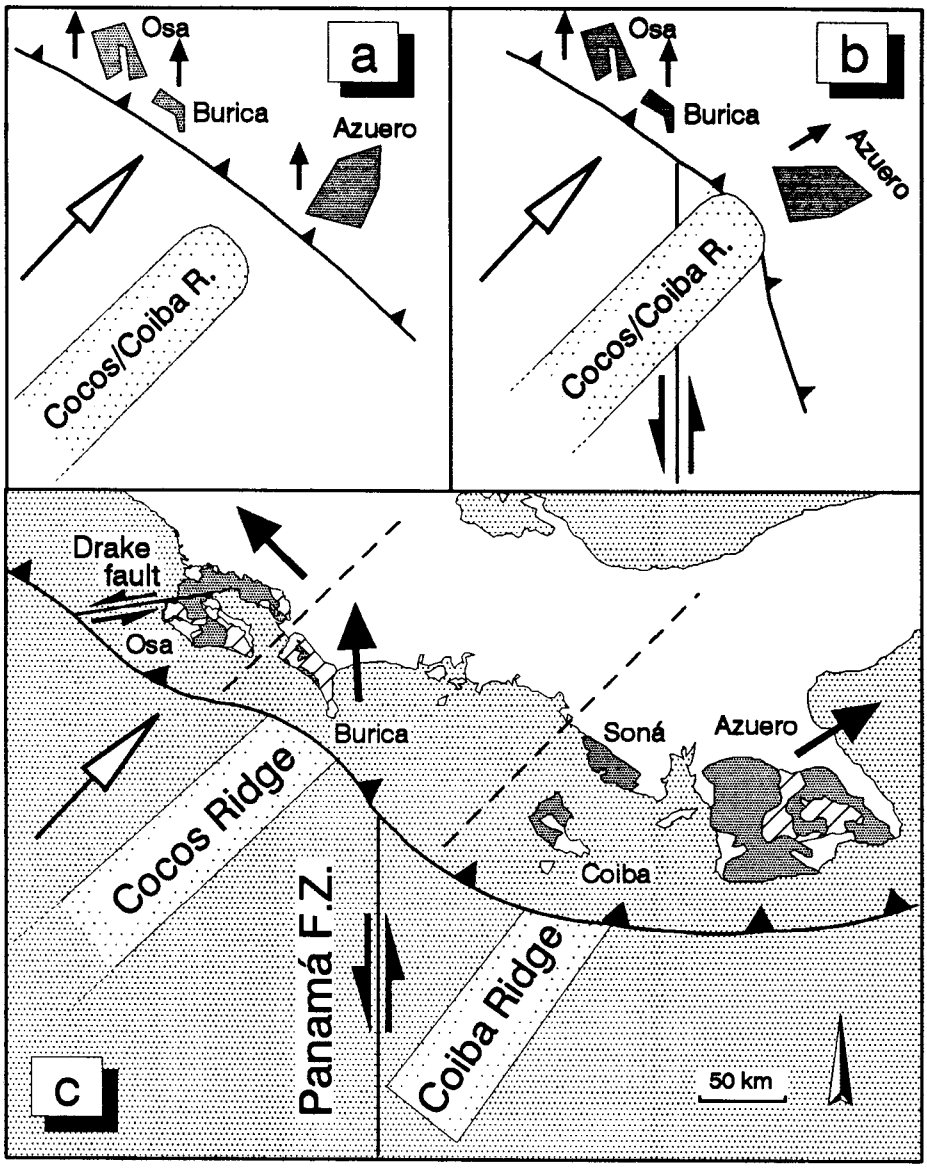


Figure 12. Sketch map to explain block rotations in southeastern Costa Rica and western Panama by the Vogt Hypothesis (modified after McCabe, 1984). Panama Fracture Zone (F.Z.) after Berrangé and Thorpe (1988). Dotted areas, ophiolitic rocks. Hatched areas, sedimentary cover. Bold arrows show direction of declination of ChRM of the respective area. Open arrows show motion of Cocos plate. (a) Early Miocene; (b) late Miocene; (c) Pliocene to Present.

experienced diverging rotation. The fragmentation and individual block rotation are documented by the paleomagnetic results.

We consider the collision of the Cocos Ridge with the Central American landbridge (Caribbean plate) responsible for the formation of the Costa Rica Fracture Zone (Fig. 1). This sinistral fault was detected in seismic profiles by Crowe and Buffler (1983). Another sinistral fault that parallels the Costa Rica Fracture Zone is the Drake fault north of the Osa peninsula (Fig. 12c; Meschede and others, 1988).

CONCLUSIONS

Our model for the plate-tectonic evolution of Central America is based mainly on the paleomagnetic study of the Central American ophiolite complexes and associated sedimentary sequences, and the magmatic evolution of basaltic rocks. There is no compelling argument in favor of the theory that the ophiolitic basement originated in a distant place in the open Pacific. On the contrary, the paleomagnetic results support an approximately *in situ* formation of the oceanic basement and a paleogeographic behavior comparable to that of the South American continent. The oceanic crust is therefore considered to have been connected to South America until it carried out some lateral shift along a transform fault since the terminating Cretaceous epoch.

The stages of evolution of the Central American ophiolite complexes are as follows. (1) Oceanic crust formed during Jurassic time as a product of the newly formed Caribbean spreading center. (2) In Late Cretaceous time, primitive intraplate magmatism thickened the oceanic basement, forming an oceanic plateau in the Caribbean region ("sill event"). On the western margin of the plateau, the magmatic intraplate accretion was augmented by primitive island-arc magmatism. Complex tectonic interaction between the stiffened western margin of the plateau and adjacent continental blocks caused nappe formation in northwestern Costa Rica. (3) The Miocene collision of the Cocos Ridge with the Caribbean plate margin caused fragmentation and block rotation in the region southeast of the Costa Rica Fracture Zone.

ACKNOWLEDGMENTS

The German Academic Exchange Service (DAAD) and its former coordinator for Central America, P. Sprechmann, gave valuable and multifaceted support for the investigations in Costa Rica. Financial support was also given by the German Science Foundation (DFG). Logistic help was given by Escuela de Geología, University of Costa Rica in San José, and RECOPE, the Costa Rican oil company. H. Seyfried, formerly of San José, helped with the organization of field work and gave a lot of information concerning the geology of the region. The paleomagnetic work was carried out in the Paleomagnetic Laboratory, Mining University of Leoben, Austria, in cooperation with H. Mauritsch, who also contributed during many valuable discussions. Critical reviews by J. A. Tarduno and F. Heller, Zürich; H. Soffel, Munich; D. Appel and L. Ratschbacher, Tübingen; as well as T. Donnelly, R. Speed, W. Gose, L. Glover, W. D. MacDonald, and A. G. Sylvester as reviewers considerably improved the manuscript. All of this is gratefully acknowledged.

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MANUSCRIPT RECEIVED BY THE SOCIETY MARCH 19, 1991
REVISED MANUSCRIPT RECEIVED FEBRUARY 11, 1992
MANUSCRIPT ACCEPTED FEBRUARY 26, 1992

Printed in U.S.A.